

Challenge 18: Municipal Solid Waste & Circular Economy Agent

Domain: Urban Waste Management

The Challenge

Rapid urbanization across Gujarat's municipal corporations has outpaced door-to-door segregation compliance and collection-route efficiency in many wards, creating a gap between mandatory segregation policies and ground-level implementation.

Objective

Create Agentic AI solutions that optimize waste-collection routes dynamically, improve segregation compliance through citizen nudges, automatically route citizen grievances to the appropriate municipal desk, and provide ward-level analytics for better decision-making.

Suggested AI Agents

- Route Optimization Agent
- Segregation Compliance Tracking Agent
- Grievance Intake Agent (Multilingual Chat/Voice)
- Municipal Routing Agent
- Ward Analytics Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM Cloud